

Literature Status for Two Questions in arXiv:1905.07201

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Status

This note records that two open-problem signals in Albiac–Ansorena–Cudeiro–Doucha, arXiv:1905.07201, are settled by the later paper Albiac–Ansorena–Cuth–Doucha, arXiv:2006.08018.

The Schauder-basis question is explicitly answered by the later paper. The Euclidean subset complementability question is answered by direct specialization of a later complementability result.

Source Questions

In Section 6 of arXiv:1905.07201, the authors ask:

Does $\mathcal{F}_p([0, 1]^d)$ have a Schauder basis for each $d \in \mathbb{N}$ and each $p \in (0, 1]$?

The later paper arXiv:2006.08018 identifies this as Question 6.5 of arXiv:1905.07201.

The same source section also asks:

Let $\mathcal{M} \subset \mathbb{R}^d$ and $p \in (0, 1]$. Is $\mathcal{F}_p(\mathcal{M})$ isomorphic to a complemented subspace of $\mathcal{F}_p(\mathbb{R}^d)$?

Later Answer

The abstract of arXiv:2006.08018 states that the authors prove $\mathcal{F}_p(\mathbb{R}^d)$ admits a Schauder basis for every $p \in (0, 1]$, and that they give explicit formulas for bases of both $\mathcal{F}_p(\mathbb{R}^d)$ and its isomorphic space $\mathcal{F}_p([0, 1]^d)$. The introduction then says explicitly that this answers arXiv:1905.07201, Question 6.5, in the positive.

The theorem-level statement is arXiv:2006.08018, Theorem 2.9:

for every $0 < p \leq 1$ and $d \in \mathbb{N}$, $\mathcal{F}_p([0, 1]^d)$ has a Schauder basis.

This directly resolves the basis question.

For the complementability question, the same later paper records that for every $d \in \mathbb{N}$ there is a constant $C = C(p, d)$ such that, whenever $\mathcal{M} \subset \mathcal{N} \subset \mathbb{R}^d$, the space $\mathcal{F}_p(\mathcal{M})$ is C -complemented in $\mathcal{F}_p(\mathcal{N})$. Taking $\mathcal{N} = \mathbb{R}^d$ yields a positive answer to the source question: $\mathcal{F}_p(\mathcal{M})$ is isomorphic to a complemented subspace of $\mathcal{F}_p(\mathbb{R}^d)$ for every $\mathcal{M} \subset \mathbb{R}^d$ and $p \in (0, 1]$.

Scope

No new theorem is claimed in this packet. The purpose is only to clear these two target-queue signals and record the exact later source that answers them. The neighboring questions in arXiv:1905.07201 about the metric approximation property, isometric copies of ℓ_p , and metric quotients are not resolved here.

References

1. F. Albiac, J. L. Ansorena, J. Cudeiro, and M. Doucha, *Embeddability of ℓ_p and bases in Lipschitz free p -spaces for $0 < p \leq 1$* , arXiv:1905.07201.
2. F. Albiac, J. L. Ansorena, M. Cuth, and M. Doucha, *Structure of the Lipschitz free p -spaces $\mathcal{F}_p(\mathbb{Z}^d)$ and $\mathcal{F}_p(\mathbb{R}^d)$ for $0 < p \leq 1$* , arXiv:2006.08018.